

Denoising PSRs with CNNs

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1 Introduction

Denoising lunar images is essential for reliable interpretation of permanently shadowed regions, where high noise levels can hide small craters, local albedo changes, and regolith texture. In this open project, the task was to adapt the previous denoising pipeline to a harder dataset of higher-resolution, noisier lunar images. The training set contains paired noisy and clean images, while the test set contains only noisy images with clean targets held back. We therefore treated the task as supervised image-to-image regression during training, selected models using paired validation metrics, and inspected blind-test predictions qualitatively.

Starting from the Task 2 baseline, we investigated changes in architecture, capacity, optimization, loss functions, regularization, and output activation. The best model was a large four-stage Deep U-Net. It achieved the best completed validation score: $\text{MSE} = 0.001758$, $\text{MAE} = 0.03213$, and $\text{PSNR} = 27.55$ dB.

2 Method

We explored several architecture families: standard U-Nets, attention-gated U-Nets, residual CNN denoisers, residual U-Nets, autoencoders, DINOv2-based feature decoders, and conditional diffusion models. U-Net-based models were especially well matched to this task because the encoder-decoder structure captures multi-scale noise patterns, while skip connections preserve high-resolution spatial details. Attention U-Nets were tested to suppress irrelevant skip features, and ResNet-style denoisers were tested as a full-resolution alternative. Diffusion models and DINOv2 decoders were included as higher-risk approaches to test whether more advanced representation learning could improve perceptual quality.

We first ran ablation studies to reduce the search space. The ablations varied feature-channel capacity, architecture depth, loss function, optimizer, learning rate, output activation, and regularization. After identifying Deep U-Nets as the strongest family, we used Optuna for hyperparameter optimization. Optuna samples configurations from a defined search space, trains each trial for a limited budget, evaluates validation PSNR, and uses pruning to stop trials whose intermediate score is unlikely to become competitive. For the Deep U-Net search, the sampled parameters included feature scale, dropout, sigmoid versus tanh output, learning rate, weight decay, optimizer, and batch size.

2.1 Dataset and Metrics

The dataset consists of 19 500 paired noisy/clean lunar images of size 128×128 and 1 000 blind noisy test images. Arrays were converted to `float32`, normalized to $[0, 1]$ by dividing by 255 when necessary, and loaded as single-channel tensors. Final reruns used a deterministic 5% validation split and were trained on Euler GPUs.

On paired validation data, PSNR was computed in the standard full-reference way from the MSE between the denoised output $\hat{x}$ and the clean target x :

$$\text{PSNR} = 10 \log_{10} \left(\frac{1}{\text{MSE}(\hat{x}, x)} \right),$$

because images are normalized to $[0, 1]$. On the blind test set, clean targets are not available, so true test MSE, MAE, or PSNR cannot be computed locally. The blind test images were therefore denoised for submission and inspected visually, while all quantitative model selection was based on paired validation data.

2.2 Final Model Architecture

Only the final model is shown in detailed tabular form. The selected architecture is a large four-stage Deep U-Net with feature channels $[64, 128, 256, 512]$, a 1024-channel bottleneck, batch normalization, ReLU activa-

tions, dropout $p = 0.10887$, transposed-convolution upsampling, concatenated skip connections, and a sigmoid output.

Table 1. Detailed architecture of the best model, a large four-stage Deep U-Net. Input and output images have shape (1, 128, 128).

Stage	Layer Name	Operation / Output Shape	Parameters
Encoder	input	Grayscale input (1, 128, 128)	0
	enc1	DoubleConv (64, 128, 128)	37,696
	enc2	MaxPool + DoubleConv (128, 64, 64)	221,696
	enc3	MaxPool + DoubleConv (256, 32, 32)	885,760
	enc4	MaxPool + DoubleConv (512, 16, 16)	3,540,992
Bottleneck	bottleneck	MaxPool + DoubleConv (1024, 8, 8)	14,159,872
Decoder	up4	ConvTranspose2D (512, 16, 16)	2,097,664
	dec4	Concatenate skip + DoubleConv (512, 16, 16)	7,079,936
	up3	ConvTranspose2D (256, 32, 32)	524,544
	dec3	Concatenate skip + DoubleConv (256, 32, 32)	1,770,496
	up2	ConvTranspose2D (128, 64, 64)	131,200
	dec2	Concatenate skip + DoubleConv (128, 64, 64)	442,880
	up1	ConvTranspose2D (64, 128, 128)	32,832
	dec1	Concatenate skip + DoubleConv (64, 128, 128)	110,848
Output	out_conv	1×1 Conv2D + Sigmoid (1, 128, 128)	65
Total Trainable Parameters			31,036,481

2.3 Training Setup

The final model was trained with MSE loss, AdamW, learning rate $9.664 \cdot 10^{-4}$, weight decay $2.472 \cdot 10^{-3}$, batch size 16, and a cosine learning-rate scheduler for 150 epochs. Dropout, AdamW weight decay, validation checkpointing, and the U-Net skip-connection structure were used to control overfitting. The best checkpoint was selected by validation error. For comparison, the best Attention U-Net candidates used a combined $0.5 \cdot \text{MSE} + 0.5 \cdot \text{L1}$ loss and learning rates around $4 \cdot 10^{-4}$, while diffusion candidates used MSE loss, AdamW, and DDIM-style validation.

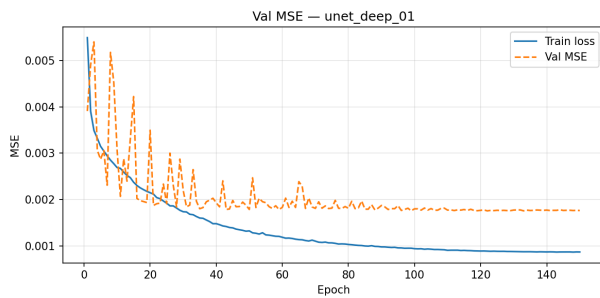
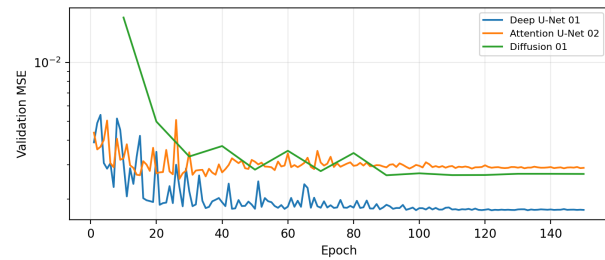
3 Results

3.1 Ablation Results

Throughout this project, a broad set of configurations was explored. The main ablation categories are summarized in Table 2. The strongest trend was architectural: deeper multi-scale U-Nets consistently outperformed flat ResNets, autoencoders, and attention-gated variants on validation MSE/PSNR. MSE was selected for the final Deep U-Net because it directly optimizes the reported pixel-wise error and PSNR. L1 or mixed losses sometimes produced visually plausible outputs, but did not improve the final validation score.

Table 2. Ablation studies overview. The best performing choices for the final model family are highlighted in bold.

Ablation Study	Parameter Varied	Candidates / Configurations
Network Capacity	Feature Channels	Small, default, and large feature maps; ResNet variants with 8 and 16 residual blocks
Architecture	Structural Block / Gating	Standard U-Net, Attention U-Net, Deep U-Net (4 stages) , Res-UNet, Autoencoder, Flat ResNet, diffusion, DINOv2 decoder
Loss Function	Optimization Objective	MSE , L1, combined MSE/L1, MS-SSIM/L1-inspired loss
Optimization Setup	Optimizer & Learning Rate	Adam and AdamW ; Optuna-searched learning rates and weight decay
Output Layer	Activation Function	Sigmoid vs. tanh mapped to $[0, 1]$

**(a)** Final Deep U-Net training loss and validation MSE.**(b)** Validation MSE for representative final configurations.**Fig. 1.** Loss curves used to assess training stability and model selection.

3.2 Final Results

The best model was a large four-stage Deep U-Net. It reached a best validation MSE of 0.001758, MAE of 0.03213, and PSNR of 27.55 dB. A closely related Deep U-Net configuration reached PSNR 27.53 dB, indicating that the Deep U-Net family was robust rather than a single lucky configuration.

Table 3. Final candidate comparison on paired validation data.

Model	Main difference	Best val. MSE	Val. PSNR [dB]
Best Deep U-Net	large 4-stage U-Net, MSE	0.001758	27.55
Alternative Deep U-Net	higher LR and dropout	0.001767	27.53
Attention U-Net	attention gates, combined loss	0.002542	25.95
Diffusion model	conditional DDPM-style model	0.002654	25.76

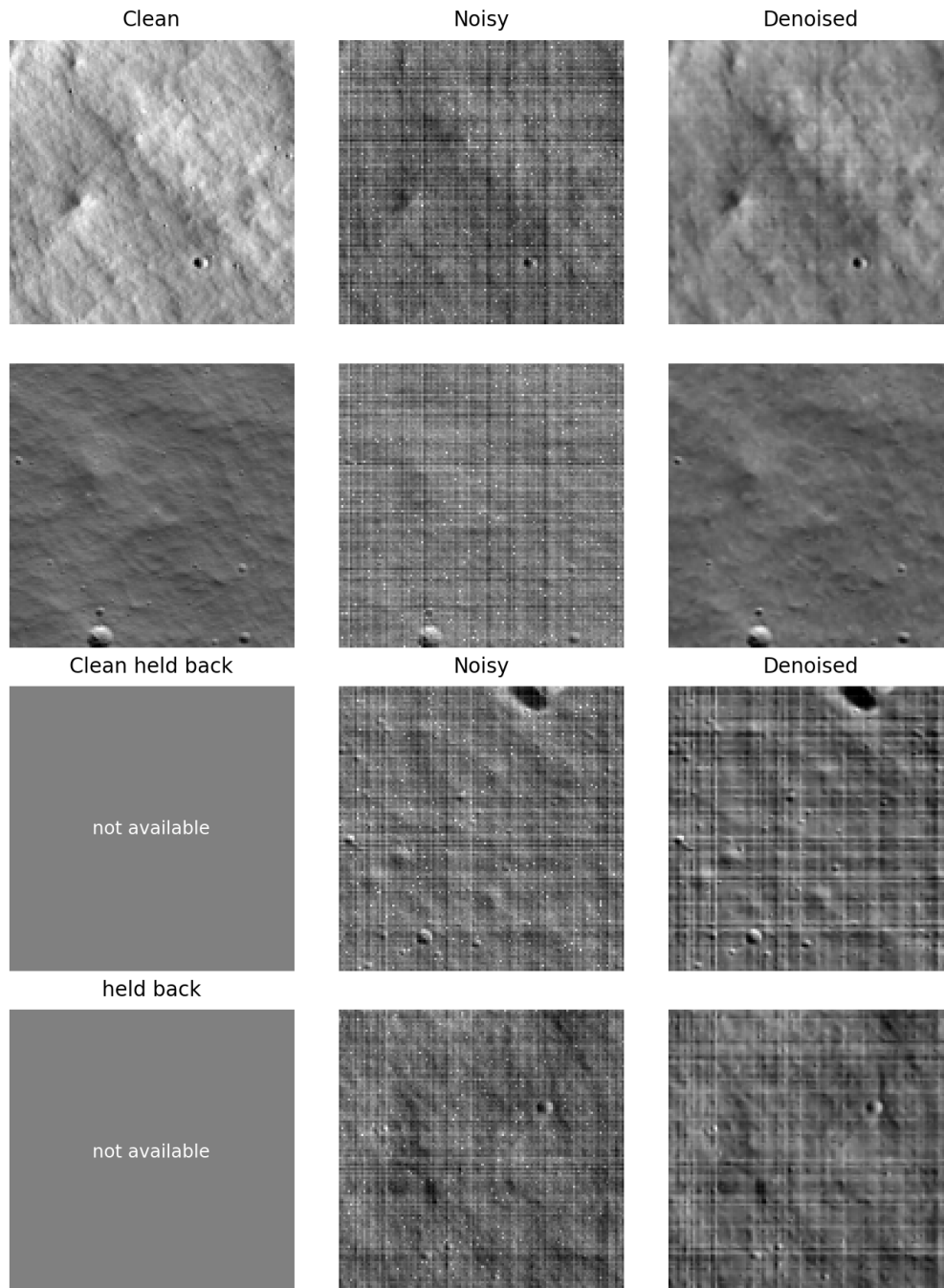


Fig. 2. Two paired training examples and two blind-test examples. For blind-test images, the clean targets are held back by the assignment. The final Deep U-Net removes most high-frequency noise while preserving the dominant crater structures; some very fine contrast is still smoothed.

WE NEED TO ADD THE TWO BLIND TEST IMAGES (old one were shit)

4 Discussion

4.1 Challenges

The main challenge was balancing denoising strength against preservation of small-scale lunar texture. Larger models reduced validation MSE, but also increased the risk of overfitting during long runs. This was addressed with dropout, AdamW weight decay, cosine learning-rate scheduling, and validation checkpointing. The valida-

tion curves show small oscillations, which is expected for a relatively small validation subset, but no divergence or training instability was observed.

Architecturally, the flat ResNet and autoencoder baselines were less competitive because they lacked the same combination of multi-scale context and direct high-resolution skip features. Attention U-Nets were more expressive, but in this problem the gates did not improve the final score and may have suppressed faint but useful surface texture. Diffusion models were a valuable risk-taking direction and produced plausible denoised images, but validation and inference were slower and the final quantitative metrics remained below the feed-forward Deep U-Net.

The blind test set has no clean target, so true test MSE/PSNR cannot be computed locally. For this reason, final model selection was based on paired validation metrics and qualitative inspection of the denoised blind-test predictions.

5 Conclusion

The final submitted model is a large four-stage Deep U-Net trained with MSE and AdamW. It was selected because it achieved the best completed validation score, trained stably on Euler, and gave the best visual trade-off between removing noise and preserving lunar structure. The project also showed that systematic ablations followed by Optuna refinement were more effective than trying isolated complex models: the ablations identified Deep U-Nets as the strongest family, and Optuna refined the learning rate, regularization, activation, and batch size within that family.

Use of Generative AI

Generative AI tools including ChatGPT-5.3, Copilot, and Claude were used during the preparation of this document for LaTeX formatting, drafting of text and coding. The final content has been carefully reviewed and validated by the authors.